

Anthony Schurle

anthonyschurle.com | as556@rice.edu | github.com/anthony-schurle | Houston, Texas

Education

Rice University <i>B.S. in Mathematics; B.A. in Computer Science</i> GPA: 3.80/4.00	Houston, Texas Aug. 2024 – May 2028
Northland Christian School <i>Valedictorian</i>	May 2024

Research Interests

Logic in Computer Science; Descriptive Complexity; Finite Model Theory; Structural Graph Theory

Expository Work

Tangle-tree duality in abstract separation systems <i>Supervised reading course; mentor: Prof. Illya V. Hicks, Rice University</i>	Spring 2026
- Studying Diestel–Oum’s tangle-tree duality theorem for abstract separation systems and related literature - Preparing an expository note under faculty supervision	

Teaching Experience

Incoming Teaching Assistant <i>Computational Thinking (COMP140)</i> George R. Brown School of Engineering and Computing, Rice University	Aug. 2026 – Dec. 2026
Teaching Assistant <i>Algorithmic Thinking (COMP182)</i> George R. Brown School of Engineering and Computing, Rice University	Jan. 2026 – May 2026
Teaching Assistant <i>Computational Thinking (COMP140)</i> George R. Brown School of Engineering and Computing, Rice University	Aug. 2025 – Dec. 2025

Honors & Awards

Trustee Distinguished Academic Scholarship , Rice University	Apr. 2024
President’s Honor Roll , Rice University	Fall 2025

Relevant Coursework

Graduate & Cross-Listed: General Topology; Automata, Formal Languages, and Computability
Mathematics: Abstract Algebra I; Honors Analysis; Number Theory; Graph Theory; Honors Linear Algebra; Honors ODEs
Computer Science: Reasoning About Algorithms
Independent Study: *A Mathematical Introduction To Logic* (Enderton)

Selected Computational Projects

Graft (<i>Interactive Graph Theory Sandbox</i>) <i>JavaScript, SVG</i>
- Built an interactive graph editor for constructing, modifying, and analyzing finite graphs through a visual interface - Implemented real-time computation of graph invariants, including degree sequences, connectivity, bipartiteness, and exact chromatic number for small graphs - Added export support for set notation, adjacency matrices, and LaTeX/TikZ diagrams

Epidemiological Transmission Mapping | *Python*

- Modeled bacterial infection spread as a complete weighted directed graph using genetic Hamming distance and epidemiological data
- Implemented greedy algorithms to infer a rooted directed minimum spanning tree representing likely transmission pathways

Technical Skills

Languages: Python, C, Java, JavaScript, RISC-V Assembly, x86-64 Assembly
Tools: LaTeX, Git & GitHub, Docker, Linux